

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 3.2 CURRENT ELECTRICITY – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the Huruma Estate phenomenon. Use what you learned in Lessons 1 to 12 to answer every section in the space under each prompt. For each answer, state a claim, give evidence from our investigations, and explain your reasoning using ideas such as Ohm's Law, EMF, internal resistance, series and parallel rules, power and energy. Include units in every calculation and show your working. Write in full sentences.

Section 1: Dim lights and a slow kettle after the blackout

After a Kenya Power blackout in Huruma Estate, Nairobi, the lights glowed dimly and the kettle took longer to boil when electricity was first restored. A few minutes later everything worked normally. Explain what happened. In your answer: (a) say what you measure with a voltmeter on the supply terminals when current is flowing compared with when no current flows; (b) use the idea of internal resistance and the equation for terminal voltage; (c) link the change in voltage to the power delivered to the bulbs and kettle using $P = V^2/R$.

Claim: When power was restored, many appliances switched on at once and drew a large current. This lowered the voltage reaching the sockets, so the lights received less power and the kettle heated slowly. Evidence: In Lesson 2 the voltmeter reading across a dry cell dropped below its open-circuit reading (the EMF) when we connected a load, and the drop was bigger with a smaller resistor (larger current). In Lesson 8 our V against I graph sloped downwards, with the y-intercept equal to the EMF and the negative gradient equal to the internal resistance r . Reasoning: The supply has an EMF E but also internal resistance, so terminal voltage $V = E - Ir$. A large surge current I makes the lost volts Ir large, so V at the sockets is below the normal 240 V. Power delivered to a device is $P = V^2/R$. A lower V gives a much lower P for the same resistance R . The filament therefore transfers less energy per second and glows dimly, and the kettle element transfers heat to the water more slowly. When the surge settled, the current fell, Ir fell, the voltage returned to normal and the lights brightened.

Section 2: The bulb on the long, thin extension cord

A neighbour in Huruma Estate has a bulb on a long, thin extension cord. It is always dimmer than a bulb plugged directly into the wall socket,

(a) The cord is in series with the appliances, so it takes part of the supply voltage. The table shows 31.3 V is dropped across the cord, leaving only 208.7 V (not 240 V) across the bulb and charger. $31.3 \text{ V} + 208.7 \text{ V} = 240 \text{ V}$. Less voltage across the bulb means less current and less power ($P = V^2/R$), so it glows dimmer. This is not a fault in the supply; it is how a series element behaves. (b) From $p = RA/L$, $R = \rho L/A$. Resistance increases with length ($R \propto L$) and decreases with larger cross-sectional area ($R \propto 1/A$). A long, thin cord therefore has high resistance, unlike the thick copper house wiring. In Lesson 4, thin, long wire gave much higher

even when power is fully restored. In the Huruma Extension Cord Circuit from Lesson 7, the data were:

Part of circuit	Voltage (V)
Supply	240
Across the 6 Ω extension cord	31.3
Across the parallel appliances (bulb in Socket A and charger in Socket B)	208.7

(a) Explain why the bulb is dimmer, using the data. (b) Describe how the cord's length and cross-sectional area affect its resistance, and how this links to $\rho = RA/L$. (c) Calculate the current flowing through the cord. (d) Explain why this is a problem for energy delivery and safety.

resistance than short, thick wire of the same material. (c) Using Ohm's Law on the cord: $I = V/R = 31.3 \text{ V} \div 6 \Omega \approx 5.2 \text{ A}$. (d) The cord wastes energy as heat ($P = I^2R$ and $Q = I^2Rt$), so less energy reaches the appliances. If the cord is overloaded it can become hot enough to damage the insulation and start a fire, as in the charred cord from Huruma Estate.

Section 3: Why homes are wired in parallel

Explain why all the appliances in a Kenyan home are connected in parallel to the 240 V supply rather than in series. Use what you observed in Lessons 5 and 6 (Christmas-style bulbs in series and the

Claim: Parallel wiring lets every appliance receive the full supply voltage and work independently. Evidence: In Lesson 5, when we added bulbs in series the current was the same everywhere and the bulbs got dimmer because the source voltage was shared ($V_{\text{source}} = V_1 + V_2 + V_3$) and the total resistance increased. In Lesson 6 our parallel circuits showed about 6 V across each branch, equal to the battery voltage. Unscrewing one bulb did not affect the other, and the main-line current equalled the sum of the branch currents. Reasoning: In parallel, each branch is connected directly across the supply, so each appliance gets the full 240 V and operates normally; the lights and kettle do not affect each other. If the home were in series, every extra appliance would add resistance, reduce current and make everything dimmer, and one switched-off appliance would break the whole loop. The cost

parallel circuits you built). Mention voltage, current and what happens when one appliance is switched off or removed.	of parallel wiring is that adding appliances increases total current, because equivalent resistance falls ($1/R_{eq} = 1/R_1 + 1/R_2 + \dots$). The wiring and fuse must be able to carry this larger current safely.
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Section 4: Power, energy and cost at home

A 3 kW electric jiko is used in a Nairobi home on a 240 V supply. Kenya Power charges about Ksh 12.00 per unit (1 kWh). (a) Calculate the current the jiko draws. (b) Calculate how long it takes to use 1 kWh and the cost of running it for that time. (c) Explain why a kettle that runs on lower terminal voltage during the Huruma power restoration takes longer to boil water, and say whether the energy needed to boil the same water changes.	(a) $P = IV$, so $I = P/V = 3000 \text{ W} \div 240 \text{ V} = 12.5 \text{ A}$. (b) $E = Pt$, so $t = E/P = 1 \text{ kWh} \div 3 \text{ kW} = 1/3 \text{ h} = 20 \text{ minutes}$. Cost = $1 \text{ kWh} \times \text{Ksh } 12.00 = \text{Ksh } 12.00$. (c) Lower terminal voltage reduces the current ($I = V/R$) and the power delivered ($P = V^2/R$). Less power means less energy is transferred to the water each second, so more minutes are needed to reach boiling point. Heating the same water to boiling needs about the same amount of energy, so the energy used, $E = Pt$, stays about the same: the power is lower but the time is longer. It may be slightly more because extra heat is lost to the surroundings during the longer heating time. So the slow kettle is mainly a problem of time, not a much bigger bill. The Kenya Power bill is charged in kWh.
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Section 5: Designing a safe, efficient circuit for the Wanjiku Family Home

The Wanjiku family in Githurai, Nairobi, has a 15 A mains fuse on 240 V. They want to run the 3 kW electric jiko and a second 3 kW appliance at the same time on this fuse. (a) Calculate the total current if both run together and decide whether the 15 A fuse is safe. (b) Explain what the fuse does and what could happen if a much bigger fuse or a thin extension cord were used instead. (c) Give three design rules, with reasons, for wiring the home so that every appliance gets the right amount of energy safely and efficiently. Answer the driving question:	(a) Each 3 kW appliance draws $I = P/V = 3000 \div 240 = 12.5 \text{ A}$. In parallel the currents add: $12.5 \text{ A} + 12.5 \text{ A} = 25 \text{ A}$. This is above 15 A, so the fuse would melt and break the circuit. This is correct and safe: the fuse is protecting the wiring. The two appliances should not run together on this circuit. (b) A fuse melts when the current is too high and breaks the circuit before the wires overheat. A fuse with too high a rating would not melt, so the wires could carry more current than they are meant to ($Q = I^2Rt$) and could start a fire. A thin extension cord has high resistance, wastes energy as heat and can become a fire hazard. (c) Rule 1: wire appliances in parallel so each gets the full 240 V and works independently. Rule 2: use thick copper cables and short cords, because low resistance ($R = \rho L/A$) reduces voltage drop ($V = IR$) and heat loss ($P = I^2R$). Rule 3: choose a fuse (or circuit breaker) rated to suit the total current and earth metal-cased appliances, so overcurrent or fault current is stopped safely. Answer to the driving question: devices get less energy than expected when voltage is lost inside the supply ($V = E - Ir$) or across high-resistance wires in series, which reduces power ($P = V^2/R$). Good design keeps wire resistance low, uses parallel connections, matches fuse ratings to current, and uses energy ($E = Pt$) within the monthly budget.
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why do devices sometimes get less energy than expected, and how do we design circuits to prevent this?	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning	Gives a clear claim for each phenomenon, supports it with specific data from our investigations (for example 31.3 V and 208.7 V), and links evidence to the claim with correct scientific reasoning.	Gives claims and some evidence from the lessons, but reasoning is partly missing or does not fully link the evidence to the claim.	Gives statements with little or no evidence, or evidence that does not match the claim.
Use of circuit ideas ($V = IR$, internal resistance, series and parallel rules)	Uses $V = IR$, $V = E - Ir$, and the series and parallel rules correctly and consistently to explain both the post-blackout dimming and the extension cord.	Uses most of the ideas correctly, with a few minor errors or missing links.	Uses ideas incorrectly or confuses series with parallel, or EMF with terminal voltage.
Calculations of power, energy, current and cost	Calculations are correct with formula, substitution, working and units (A, W, kWh, Ksh).	Method is correct, but there are small arithmetic or unit errors.	Method is incorrect or incomplete, or units are missing.
Safe and efficient design decisions	Justifies parallel wiring, thick cables, fuse choice and earthing, and links each to the driving question about energy delivery and safety.	Gives design rules with partial justification.	Gives design rules without reasons, or rules that are unsafe.
Communication and Kenyan context	Writing is clear and well organised, uses correct terms, and refers to the Huruma Estate and Kenya Power contexts.	Writing is mostly clear, with some terms misused or limited reference to context.	Writing is hard to follow, with little use of scientific terms or context.

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